

Consistency Questionnaire Results

Q1. Missing physiological attributes

■ No inconsistencies found.

Q2. Multiple HR per ActorState

■ No inconsistencies found.

Q3. Multiple HRV per ActorState

■ No inconsistencies found.

Q4. Multiple RR per ActorState

■ No inconsistencies found.

Q5. Multiple SpO2 per ActorState

■ No inconsistencies found.

Q6. Multiple Drowsiness per ActorState

■ No inconsistencies found.

Q7. Missing high-level attributes

■ 7 inconsistencies found.

Running Q7. Missing high-level attributes...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_de743f64bfb645ab9d0faa4287219144'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_33d29622dbae4cc49c776d821075a37a'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_7b941b49cadb4f70aa087131d49472a8'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_6458c6a80dfd4896a0db1a5eea9fdf2a'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_3e20688d84644bed95e3f68467159d79'),)
```

Q8. Multiple Fatigue per ActorState

■ No inconsistencies found.

Q9. Multiple Attention per ActorState

■ No inconsistencies found.

Q10. Multiple Unresponsiveness per ActorState

■ No inconsistencies found.

Q11. Multiple EyeState per ActorState

■ No inconsistencies found.

Q12. Multiple MouthState per ActorState

■ No inconsistencies found.

Q13. Multiple Age or Sex per ActorState

■ No inconsistencies found.

Q14. Multiple Accessories/Demographic/FaceCharacteristics per ActorState

■ No inconsistencies found.

Q15-A. HR instances classified into more than one HR-level class

■ No inconsistencies found.

Q15-B. HRV instances classified into more than one HRV-level class

■ No inconsistencies found.

Q15-C. RR instances classified into more than one RR-level class

■ No inconsistencies found.

Q15-D. SpO2 instances classified into more than one SpO2-level class

■ No inconsistencies found.

Q15-E. Drowsiness instances classified into more than one Drowsiness-level class

■ No inconsistencies found.

Q16-A. HR without any HR-level classification

■ No inconsistencies found.

Q16-B. HRV without any HR-level classification

■ No inconsistencies found.

Q16-C. RR without any RR-level classification

■ No inconsistencies found.

Q16-D. SpO2 without any SpO2-level classification

■ No inconsistencies found.

Q16-E. Drowsiness without any Drowsiness-level classification

■ No inconsistencies found.

Q17. HRV instances with >1 HRV level

■ No inconsistencies found.

Q18. RR instances with >1 RR level

■ No inconsistencies found.

Q19. SpO2 instances with >1 SpO2 level

■ No inconsistencies found.

Q20. Drowsiness instances with >1 Drowsiness level

■ No inconsistencies found.

Q21. Violation "Sleep=> Inattentive & Unresponsive"

■ No inconsistencies found.

Q22. Violation "Drowsy & attentive=> at-risk unresponsiveness"

■ No inconsistencies found.

Q23. Violation "Sleep or unresponsiveness => eyes closed & mouth open"

■ 4 inconsistencies found.

```
Running Q23. Violation "Sleep or unresponsiveness => eyes closed & mouth open"...
■ Inconsistencies found (2 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_33d29622dbae4cc49c7
76d821075a37a'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#undefinedstate_instance'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#unresponsive_instance'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#eye_state_2024_4_15_0_16_33'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#closedstate_instance'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#mouth_state_2024_4_15_0_16_33'))
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_3e20688d84644bed95e
3f68467159d79'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#undefinedstate_instance'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#unresponsive_instance'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#eye_state_2024_4_15_0_35_56'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#closedstate_instance'),
rdfLib.term.URIRef('http://test.org/in_cabin_domain#mouth_state_2024_4_15_0_35_56'))
```

Q24. Is any state missing its threshold profile?

■ No inconsistencies found.

Q25. Are there any threshold profiles not used by an Actor State?

■ 7 inconsistencies found.

```
Running Q25. Are there any threshold profiles not used by an Actor State?...
■ Inconsistencies found (17 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
```

Q26. Detect future categories accidentally left unmapped

■ No inconsistencies found.

Q27. Show if new data falls outside modeled thresholds.

■ No inconsistencies found.

Q28. Check if new states appear without corresponding GCI or SWRL effects.

■ No inconsistencies found.

Q29. Future adaptability often breaks.

■ 7 inconsistencies found.

```
Running Q29. Future adaptability often breaks....
■ Inconsistencies found (18 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
```

Q30. Unexpected extensions

■ 7 inconsistencies found.

```
Running Q30. Unexpected extensions....
■ Inconsistencies found (2334 rows)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#Ontology'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
```

Q31. Severe drowsiness (Level 9) must imply Imminent or Unresponsive

■ No inconsistencies found.

Q32. Moderate drowsiness (Level 5) must be classified as DrowsinessSuspected fatigue

■ No inconsistencies found.

Q33. Low drowsiness (Level 3) must be classified as Awake fatigue

■ No inconsistencies found.

Q34. Low drowsiness (Level 3) must imply Attentive attention

■ No inconsistencies found.

Q35. High drowsiness (Level 7 or 9) must imply Inattentive attention

■ 4 inconsistencies found.

```
Running Q35. High drowsiness (Level 7 or 9) must imply Inattentive attention...
■ Inconsistencies found (2 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_33d29622dbae4cc49c776d821075a37a'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_3e20688d84644bed95e3f68467159d79'),)
```

Q36. Sleep fatigue must have both eye and mouth state defined

■ No inconsistencies found.

Q37. Awake fatigue must be Responsive and Attentive

■ No inconsistencies found.

Q38. High-risk states (Sleep or Unresponsive) must have at least one Label

■ 4 inconsistencies found.

```
Running Q38. High-risk states (Sleep or Unresponsive) must have at least one
Label...
■ Inconsistencies found (2 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_33d29622dbae4cc49c776d821075a37a'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_3e20688d84644bed95e3f68467159d79'),)
```

Q39. Every Label must target at least one ActorState

■ 7 inconsistencies found.

```
Running Q39. Every Label must target at least one ActorState...
■ Inconsistencies found (5 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_1_7'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_16_33'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_24_38'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_28_49'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_35_56'),)
```